Bacteria

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Identity Sheet

Property	Value
Name	Bacteria (Bacteriuria)
Other names	Urinary bacteria, bacteriuria, urinary tract pathogens
Nature	Prokaryotic microorganisms (not a single molecule)
Size	Typically 0.5–5 µm (cocci ~1 µm; rods 1–3 µm length)
Common species in UTI	*Escherichia coli* (75–90%), *Klebsiella pneumoniae*, *Proteus mirabilis*, *Enterococcus faecalis*, *Staphylococcus saprophyticus*
Diagnostic threshold	>=10^5 CFU/mL (significant bacteriuria); >=10^3 CFU/mL in symptomatic women

Representative morphologies:

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Cocci (spheres): o o o (e.g. Staphylococcus, Enterococcus)
Bacilli (rods): [===] (e.g. E. coli, Klebsiella, Proteus)
Chains: o-o-o-o (e.g. Streptococcus)
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Bacteria in urine represent a biomarker category rather than a single chemical entity. The presence and quantity of bacteria in urine (bacteriuria) is a key diagnostic marker for urinary tract infections (UTIs). Bacteria are detected by growth (culture), biochemical by-products (nitrite from nitrate reduction), enzyme activity (catalase, oxidase), or direct visualisation/counting.

Molecules Not to Be Confused With

Entity	Description	Key Difference
Yeast / Fungi	*Candida* spp., budding oval cells 3–8 µm	Eukaryotic; larger; may indicate fungal UTI or contamination
White blood cells (WBCs)	Leukocytes, 10–15 µm	Much larger; indicate host immune response, not organisms
Epithelial cells	Squamous cells, 40–60 µm	Very large, flat; indicate contamination if numerous
Amorphous debris	Precipitated salts/crystals	Non-biological; no metabolic activity

Medical Information

Origin

Endogenous The urinary tract is normally **sterile** above the distal urethra. The distal urethra harbours commensal flora (lactobacilli, coagulase-negative staphylococci, corynebacteria). Bacteriuria arises when: - Periurethral or-

ganisms ascend into the bladder (most common route, especially in women) - Haematogenous seeding (rare; e.g., *Staphylococcus aureus* bacteraemia) - Lymphatic spread (rare) - Direct inoculation (catheterisation, instrumentation)

The most common pathogen, *E. coli*, originates from the patient's own gastrointestinal flora.

Exogenous

- **Healthcare-associated UTIs:** Catheter-associated UTI (CAUTI) from hospital environment organisms (multidrug resistant *Klebsiella*, *Pseudomonas*, *Enterobacter*)
- **Sexually transmitted pathogens:** *Chlamydia trachomatis*, *Neisseria gonorrhoeae* (urethritis, not classic UTI)
- **Environmental waterborne organisms:** Rare (e.g., *Leptospira*)

Primary & Secondary Biological Roles

Primary role (in host context): - **Pathogen indicator:** Bacteria in urine above threshold concentrations indicate infection requiring treatment.

Secondary roles: - **Urinary microbiome research:** Emerging evidence of a low-biomass urinary microbiome even in healthy individuals (detected by enhanced culture and metagenomics), though clinical significance is uncertain. - **Nitrite production:** Gram-negative bacteria reduce urinary nitrate to nitrite — the basis for the dipstick nitrite test.

Catabolism and Elimination Pathway

- The healthy urinary tract eliminates bacteria through **voiding** (mechanical washout), **antimicrobial peptides** (defensins, cathelicidins), **urothelial mucus layer** (GAG layer), **Tamm-Horsfall protein** (uromodulin), and **immune surveillance** (neutrophils, macrophages).
- In infection, the host mounts an inflammatory response: neutrophil influx (pyuria), cytokine release, complement activation.
- Antibiotics are the primary therapeutic elimination pathway.

Expression in Humans

Normal Levels

Compartment	Reference Range
Mid-stream urine (MSU)	<10^3 CFU/mL (culture-negative / insignificant)
Catheter specimen	<10^2 CFU/mL considered negative
Suprapubic aspirate	Any growth is significant

Factors Influencing Levels

Increased bacterial count: - Female sex (short urethra) - Sexual activity - Urinary catheterisation - Urinary obstruction (stones, BPH) - Diabetes mellitus - Immunosuppression - Pregnancy - Poor hygiene / contaminated collection

Decreased / absent: - Recent antibiotic therapy - High urine flow rate (dilution) - Acidic urine pH - Adequate voiding habits - Antimicrobial properties of urine (urea, organic acids)

Associated Pathologies

Condition	Bacterial Level	Key Symptoms
Uncomplicated cystitis	>=10^3 CFU/mL (*E. coli*) in symptomatic women	Dysuria, frequency, urgency; prevalence ~50–60% of women lifetime
Pyelonephritis	>=10^5 CFU/mL + pyuria	Fever, flank pain, nausea, costovertebral angle tenderness
Asymptomatic bacteriuria (ASB)	>=10^5 CFU/mL, no symptoms	No symptoms; significant in pregnancy (~2–10% prevalence) and pre-surgery
Catheter-associated UTI (CAUTI)	>=10^3 CFU/mL + symptoms	Fever, suprapubic pain; ~80% of nosocomial UTIs
Urosepsis	Positive blood + urine culture	Systemic sepsis from urinary source; ~25% of all sepsis

Presence in Urine

Should it be normally present? No — the bladder urine is normally sterile. Low-level commensal colonisation of the distal urethra means very small numbers (<10^3 CFU/mL) may appear in voided specimens without indicating infection.

Normal levels: <10^3 CFU/mL in clean-catch midstream urine.

Form in urine: Intact, viable (or non-viable) bacterial cells — planktonic (free-floating) or occasionally in biofilms (catheter surfaces).

Pathological significance:

Finding	Possible Causes	Prevalence
>=10^5 CFU/mL	UTI (cystitis, pyelonephritis), ASB	UTI: ~150 million cases/year globally
10^{3-10}5 CFU/mL	Early/treated UTI, contamination, urethritis	Context-dependent
<10^3 CFU/mL	Normal / contamination	—

Solubility: Not applicable (cellular organisms, not dissolved molecules).

Detection in Urine

Available Clinical Assays

1. **Urine culture (gold standard):**
   - **Principle:** Urine plated on agar media (blood agar, MacConkey, CLED); incubated 18–24 h at 37 °C; colonies counted and identified.
   - **Detection:** Colony-forming units per mL (CFU/mL).
   - **LOD:** ~10^2 CFU/mL (standard inoculum loop technique).
   - **Advantages:** Definitive identification + antibiotic susceptibility testing.
   - **Disadvantages:** 18–48 h turnaround; labour-intensive.
2. **Dipstick nitrite test:**

- **Principle:** Griess reaction — bacterial nitrate reductase converts urinary nitrate to nitrite; nitrite reacts with aromatic amine + coupling agent → pink azo dye.
- **Detection:** Visual colour change on reagent pad.
- **LOD:** ~10^5 CFU/mL (requires sufficient bacteria and incubation time in bladder).
- **Sensitivity:** 45–60%; **Specificity:** 85–98%.
- **Advantages:** Rapid (<2 min), inexpensive.
- **Disadvantages:** Misses non-nitrate-reducing organisms (Enterococcus, Staphylococcus); requires bladder dwell time >=4 h.

3. **Dipstick leukocyte esterase (LE):**
   - **Principle:** Detects esterase from neutrophils (indirect marker of infection).
   - **Detection:** Indoxyl ester substrate → indigo colour.
   - **Sensitivity:** 75–90% for UTI.
   - **Not bacteria-specific** — detects host inflammatory response.
4. **Automated flow cytometry (Sysmex UF-5000, Iris iQ200):**
   - **Principle:** Fluorescent labelling of bacterial nucleic acid + flow cytometry counting.
   - **Detection:** Bacteria/µL count; fluorescent scatterplot.
   - **LOD:** ~10–100 bacteria/µL (~10^{4-10}5/mL).
   - **Advantages:** Rapid (~1 min), automated, differentiates rods vs cocci.
   - **Disadvantages:** Cannot identify species or provide susceptibility.
5. **Gram stain microscopy:**
   - **Principle:** Centrifuged urine sediment stained with crystal violet/iodine; examined microscopically.
   - **Detection:** Visual identification of bacteria morphology and Gram reaction.
   - **LOD:** ~10^{4-10}5 CFU/mL (>=1 organism per oil-immersion field in uncentrifuged urine ≈ 10^5 CFU/mL).
   - **Advantages:** Rapid, guides empirical therapy.
   - **Disadvantages:** Operator-dependent, low sensitivity.

Optimal Urine Type for Measurement

- **Clean-catch midstream urine (MSU):** Standard for culture; reduces periurethral contamination.
- **First morning urine:** Higher bacterial concentration (longer bladder dwell time >=4 h); preferred for nitrite testing.
- **Catheter specimen:** Preferred for catheterised patients.
- **Suprapubic aspirate:** Gold standard specimen (bypasses urethra) — any growth significant.
- **Transport:** Process within **2 hours** or refrigerate (4 °C) to prevent bacterial multiplication. Boric acid preservative tubes maintain counts for up to 24 h.

Actual Gold Standard

Quantitative urine culture on standard media (blood agar + MacConkey or CLED agar) with colony counting and species identification (MALDI-TOF MS, biochemical panels, or molecular methods). Threshold for significance: >=10^5 CFU/mL (Kass criteria), though >=10^3 CFU/mL is accepted for symptomatic women with pyuria. Antibiotic susceptibility testing (AST) by disc diffusion or MIC determination is integral.

Interferences in Measurement

Interference	Effect	Mechanism
Periurethral contamination	False positive	Mixed flora from distal urethra/skin
Recent antibiotics	False negative (culture)	Suppress bacterial growth
High urine flow / dilution	False negative (nitrite, culture)	Bacteria diluted below threshold

Interference	Effect	Mechanism
Short bladder dwell time	False negative (nitrite)	Insufficient time for nitrate reduction
Non-nitrate-reducing organisms	False negative (nitrite)	Enterococcus, Staphylococcus, Candida do not produce nitrite
High ascorbic acid	False negative (dipstick)	Reduces diazo coupling reaction
Boric acid preservative	May inhibit some organisms	Bacteriostatic; excessive amounts kill bacteria

Research Detection Methods

Spectroscopy Detection (UV-Vis / NIR)

- **UV-Vis turbidimetry:** Bacterial growth measured by optical density (OD) at **600 nm** or **550 nm** — standard microbiological technique. OD600 = 0.1 ≈ 10^8 CFU/mL. LOD: ~10^6 CFU/mL. Too insensitive for direct urine use.
- **NIR spectroscopy:** Some studies use NIR (700–2500 nm) with chemometrics to classify bacterial species in pure cultures. Not validated for urine matrix. LOD: ~10^{6-10}7 CFU/mL.
- **Sample prep:** None (direct measurement of liquid).

Fluorescence Detection

- **Nucleic acid staining + flow cytometry (Sysmex UF):** Polymethine/oxazine dyes (proprietary) bind bacterial DNA/RNA. Ex 488 nm / Em 530 nm (green) and Em >600 nm (red). LOD: ~10–100 bacteria/µL.
- **FISH (Fluorescence In Situ Hybridisation):** Species-specific 16S rRNA probes labelled with fluorophores (FITC: Ex 490 / Em 520 nm; Cy3: Ex 550 / Em 570 nm). LOD: ~10^3 CFU/mL. Sample prep: fixation, hybridisation (2–4 h).
- **Calcein-AM viability staining:** Live bacteria convert non-fluorescent calcein-AM to fluorescent calcein (Ex 495 / Em 515 nm). LOD: ~10^3 CFU/mL.
- **Quantum dot conjugates:** QD-antibody conjugates for pathogen-specific fluorescent detection. Ex 350–405 nm / Em 525–655 nm (size-dependent). LOD: ~10^2 CFU/mL in buffer, ~10^3 in urine.

Raman Detection Raman spectroscopy provides species-specific fingerprints of bacteria:

Peak (cm^{-1})	Assignment
~720–730	Adenine (nucleic acids)
~780–790	Cytosine/uracil ring breathing
~1003	Phenylalanine ring breathing
~1240–1260	Amide III (proteins)
~1450	CH_2 deformation (lipids/proteins)
~1660	Amide I (proteins)

- **Conventional Raman:** Single-cell identification possible with confocal Raman (532 or 785 nm laser). LOD: single cell (but requires capture/concentration from urine).
- **SERS:** Silver/gold nanoparticles enable detection in complex matrices; LOD ~10^2 CFU/mL with SERS-based assays.
- **Sample prep:** Centrifugation/filtration to concentrate bacteria; transfer to optical substrate.

FTIR Detection Bacterial FTIR fingerprints:

Band (cm^{-1})	Assignment	Notes
~1080	PO_2^- symmetric stretch (nucleic acids, phospholipids)	Species-variable
~1240	PO_2^- asymmetric stretch	—
~1540	Amide II (proteins)	—
~1650	Amide I (proteins)	—
~2850–2960	C–H stretches (lipids)	Membrane lipids
~3300	N–H / O–H stretch (proteins, water)	—

- ATR-FTIR with multivariate analysis can classify bacterial species from pure isolates (accuracy >90%). Direct detection in urine requires concentration (filtration). LOD: ~10^5 CFU/mL without concentration.
- **Sample prep:** Centrifugation, washing, drying on ATR crystal.

Voltammetry Detection

- **Electrochemical impedance spectroscopy (EIS):** Interdigitated electrodes measure impedance changes as bacteria grow in urine. LOD: ~10^{3-10}4 CFU/mL in 1–3 h.
- **Amperometric biosensors:** Antibody- or aptamer-functionalised electrodes with enzymatic signal amplification. LOD: ~10^2 CFU/mL.
- **Differential pulse voltammetry (DPV):** Direct oxidation of bacterial components at carbon electrodes. LOD: ~10^4 CFU/mL.
- **Sample prep:** Minimal; direct urine application.

Other Detection Technologies

1. **MALDI-TOF mass spectrometry:** Direct from positive cultures for rapid species ID (<1 h after growth). Sample prep: colony pick + matrix application. LOD: requires visible colony.
2. **Molecular diagnostics (PCR/LAMP):** 16S rRNA gene PCR or species-specific primers. LOD: ~10–100 CFU/mL. Turnaround: 1–4 h. Sample prep: DNA extraction.
3. **Lateral flow immunoassays:** Antibody-based strips for specific pathogens (e.g., *E. coli*). LOD: ~10^{4-10}5 CFU/mL. No instrument needed.
4. **Microfluidic/lab-on-chip:** Integrated concentration + detection (impedance, fluorescence, or optical). LOD: ~10–10^3 CFU/mL. Sample prep: automated on-chip.
5. **ATP bioluminescence:** Bacterial ATP detected by luciferin-luciferase. LOD: ~10^{3-10}4 CFU/mL. Rapid (<5 min). Non-specific (host cell ATP interferes).

Summary Table: Detection Methods Comparison

Method	LOD	Wavelength / Key Parameter	Sample Prep	Strengths	Limitations
Urine culture (gold standard)	~10^2 CFU/mL	Colony counting, 37 °C	Plating on agar	Definitive ID + AST	18–48 h
Dipstick nitrite	~10^5 CFU/mL	Pink azo dye (visual)	None	Rapid, cheap	Low sensitivity, misses Gram+

Method	LOD	Wavelength / Key Parameter	Sample Prep	Strengths	Limitations
Dipstick LE	N/A (indirect)	Indigo colour	None	Rapid	Not bacteria-specific
Flow cytometry (Sysmex)	~10^4/mL	Ex 488 / Em 530 nm (DNA dye)	None	Automated, 1 min	No species ID
Gram stain	~10^{4-10}5 CFU/mL	Visual microscopy	Centrifugation	Morphology + Gram type	Operator-dependent
FISH	~10^3 CFU/mL	Ex/Em probe-dependent	Fixation, hybridisation	Species-specific	2–4 h, costly
Raman / SERS	Single cell (confocal); ~10^2 CFU/mL (SERS)	Ex 532/785 nm	Concentration	Species fingerprint	Specialised
ATR-FTIR	~10^5 CFU/mL	1000–3000 cm^{-1}	Centrifuge, dry	Species classification	Requires concentration
EIS (impedance)	~10^{3-10}4 CFU/mL	Impedance at 10 Hz–1 MHz	None/minimal	Rapid, label-free	Growth time needed
PCR / LAMP	~10–100 CFU/mL	Molecular	DNA extraction	Ultra-sensitive, fast	Cost, contamination risk
ATP bioluminescence	~10^{3-10}4 CFU/mL	560 nm (bioluminescence)	Lysis	Rapid (<5 min)	Non-specific

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